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# *THE INITIAL INCIDENCE OF A CARBON TAX ACROSS INCOME GROUPS*

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*Carbon taxes efficiently reduce greenhouse gas emissions, but are criticized as regressive. This paper links dynamic overlapping-generation and micro-simulation models of the United States to estimate the initial incidence of carbon taxes. We find that while carbon taxes are regressive, incidence depends much more on how carbon tax revenue is used. Recycling revenues to cut capital taxes is efficient but exacerbates regressivity. Lump sum rebates are less efficient, but much more progressive, benefitting the three lower income quintiles even when ignoring environmental benefits. A labor tax swap represents an intermediate option, as it is more progressive than a capital tax swap and more efficient than a rebate.*

*Keywords: carbon tax, distribution, incidence, tax swap, income quintiles, climate change*

*JEL Codes: H22, H23, Q52*

## I. INTRODUCTION

Many economists suggest that the introduction of a price on carbon is the most efficient way to achieve reductions in greenhouse gas emissions. There are various ways this could be achieved: through cap and trade, emissions taxes, or intensity-based performance standards. A wide variety of factors influence the relative attractiveness of these different options. Among the most important of these factors is that emissions taxes or auctioned tradable permits can generate substantial revenue. Both the efficiency and distributional consequences of carbon pricing depend crucially on how that revenue is used.

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A common attribute of both the efficiency and distributional aspects of climate policy is that these aspects unfold and change over time. For example, the effect of inefficient policies that reduce economic growth is magnified in the future when lost investment is compounded and reflected in lower economic output. Distributional effects also may change over time (Mathur and Morris, 2012) or be experienced differently by younger and older households because of different patterns of consumption, savings, and sources of income (Carbone, Morgenstern, and Williams, 2012; Blonz, Burtraw, and Walls, 2012). A central challenge of addressing climate change is that the benefits of doing so also unfold in the future. Within this historical dynamic, the decision makers shaping climate policy are always the current generations. Hence, the incidence of climate policy in the near term has disproportionate effects on the decisions of voters and policy-makers in the political process.

This paper addresses the near-term effects of climate policy design within a general equilibrium framework. A substantial literature has employed partial equilibrium methods to examine distributional outcomes, and this approach may bring a high level of resolution to the question. However, a partial equilibrium approach does not account for all the linkages in the economy that can be important, especially with respect to changes in income. The introduction of a price on carbon leads to adjustments in energy, product, and factor markets, with implied changes in income. Recycling revenue from carbon policy can also dramatically affect incomes, and in a way that is not necessarily obvious ex ante (e.g., the incidence of using carbon revenue to cut labor taxes is not just on labor).

To simultaneously achieve dynamic and general equilibrium consistency with detailed resolution including a measure of incidence before the new equilibrium is achieved, we link two new models of the U.S. economy. One is a dynamic overlapping generation (OLG) general equilibrium model that solves over a long time horizon. The second is a model of incidence that refracts changes in national income onto households according to income groups based on idiosyncratic patterns in expenditure and income.

This model linkage provides key advantages in this context. Because the OLG model is a dynamic general equilibrium model, it can estimate changes in incomes resulting from carbon pricing (and from recycling the resulting carbon tax revenue), and can pick up interactions between the policy changes and pre-existing tax distortions. And the OLG structure represents effects on capital more realistically than other commonly used approaches (such as infinitely-lived agents). Linking to the detailed incidence model provides a tight connection to the underlying data on income and expenditure patterns across income groups, and allows us to see distributional effects across income groups. To our knowledge, our work is the first to link an OLG model to a separate incidence model in order to examine the distributional effects of environmental taxation.

We use this framework to compare three policy approaches to the use of revenue that would be raised with a carbon tax. One approach uses the revenue to reduce the tax on capital income, a second approach reduces the tax on labor income, and the third delivers rebates lump sum across the economy. These policies achieve quite similar outcomes with respect to emissions, but they vary somewhat into the future because of the different growth paths that emerge over time due to changes in savings and investment.

We evaluate the outcome in the first year of the climate policy, and we find the policies have disparate effects on households. Directing revenue to reduce the capital income tax has the least effect on economic wellbeing, but it is a regressive approach that yields a wide distribution of outcomes across income quintiles. The lump sum rebate is a progressive approach that benefits a majority of households, but is the most expensive policy from the perspective of overall cost. Using revenue to reduce labor taxes lies in the middle with respect to preferences, and usually has intermediate effects in terms of impact across the income distribution.

## II. LITERATURE

Government analysis of carbon pricing during the debate over H.R. 2454 (Waxman-Markey) studied many policy features such as directing some allowance value directly to households or local distribution companies, allocation to trade exposed industries, and various specific projects. The Congressional Budget Office (CBO, 2009) found the cost per household would vary greatly with lowest income households seeing a net benefit and highest income households a net cost. Middle income households would have incurred the most significant cost as a percentage of income. In contrast, the U.S. Environmental Protection Agency (EPA, 2009) and Energy Information Administration (EIA, 2009) did not look at distribution but did illustrate how the changes in the costs of various goods and services hinge on how allowance value is allocated.

Academic studies also generally find the distributional impact of energy and environmental taxes and regulations is regressive when analyzed on the basis of annual household income because poor households spend a greater fraction of their income on energy than do wealthy households.[1] This result still holds, but is reduced when models account for the effects of a carbon tax on prices of non-energy goods (Hassett, Mathur, and Metcalf, 2009; Grainger and Kolstad, 2010).

Most studies of the incidence of carbon pricing do not account for changes in income caused by carbon pricing. Computable general equilibrium (CGE) modeling addresses that problem, calculating effects on income and interactions between markets. For example, Rausch, Metcalf, and Reilly (2011) use a CGE model to show that a portion of costs is borne by the owners of resources and capital, which lessens the regressivity of the policy. The interaction between climate policy and pre-existing distortionary taxes is particularly important, and tends to increase overall costs (Parry, Williams, and Goulder, 1999). Many of these CGE models are static, and thus can model only a long-run equilibrium, not the transition to that equilibrium.

Dynamic CGE models can look at the transition, and generally provide a more realistic treatment of effects on capital. But due to computational complexity, dynamic CGE models nearly always model only a single representative agent, and thus cannot exam-

[1] Parry et al. (2007) and Morris and Munnings (2013) provide reviews of the literature.

ine distributional effects. A handful of dynamic models include multiple agents (e.g., Jorgenson et al., 2012), and thus can examine distributional effects, but this typically requires strong assumptions about agents' preferences in order to make the problem computationally tractable.

The literature also shows the impact of carbon taxes is affected by the use of the revenues (Burtraw, Sweeney, and Walls, 2009). For example, giving the carbon asset value to industry, based on output, emissions, or some other measure, is generally inefficient and regressive, as the value flows to shareholders who are predominantly in higher-income households. Using the asset value to cut income taxes can improve the overall efficiency but also is regressive, although reducing payroll taxes would be less so. Lump sum, per capita redistribution of asset value has a progressive effect, benefiting low-income households relatively more than higher-income households (Boyce and Riddle, 2007). However, this foregoes the efficiency advantage of using revenue to reduce preexisting taxes (Dinan and Rogers, 2002; Metcalf, 2009; Parry and Williams, 2010).

## III. POLICIES

This paper looks at three different policy cases built around a carbon tax of $30 per ton of $CO_2$ $(2012), beginning in 2015 (and not anticipated prior to that date), and held constant (in real terms) thereafter. The tax applies to all fossil-fuel related $CO_2$ emissions.[2]

The differences across the three policy cases are in how the carbon tax revenue is used. Each of the first two cases uses the revenue to finance cuts in pre-existing distortionary taxes, with one case focusing on capital income tax cuts, and the other on labor income tax cuts. In each case, the cuts are assumed to be structured so that the percentage-point cut in the effective marginal tax rate (on all capital income in the capital-tax recycling case, and on all labor income in the labor-tax recycling case) is constant across income levels (which implies that all income groups get the same percentage-point cut in both marginal and average tax rates).[3] In each case, this is a one-time, permanent, unanticipated cut in the relevant tax, starting at the same time as the carbon tax.

In the third case, revenue from the carbon tax is used to provide a tax-free, lump-sum annual "rebate" payment to households, with each individual (regardless of age) receiving the same dollar amount. This rebate begins at the same time as the carbon tax, and the rebate amount remains constant (in real terms) over time and is not taxable income.

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[2] This would be most easily implemented via an upstream tax on the carbon content of all fossil fuels. It could also be implemented as a downstream tax, which would be equivalent in our model, as long as the downstream tax covers all emitters.

[3] The easiest way to implement such an across-the-board tax cut for labor income would be to cut the payroll tax rate (and more specifically the Medicare portion of the payroll tax, which applies to nearly all workers and has no income cap) and replace the lost payroll tax revenue with revenue from the carbon tax. Implementing an across-the-board cut for capital income would be more difficult, because not all capital income recipients file income tax returns (and there is no analogue to the payroll tax for capital income).

In each case, we keep the long-term level of government debt (i.e., the real value of government debt over 100 years) and the cumulative net present value of real government services and real government transfers (other than the rebate policy, if any), the same as in the benchmark (i.e., no-policy-change) case. Note that this means the time paths of real deficits, transfers, and services may change slightly from the benchmark case, but that the present value of those changes must net out to zero.[4] In the two tax-swap cases, we set the size of the tax cuts for capital or labor income to achieve the specified level of long-run debt, taking the carbon tax revenue into account. In the rebate case, we return the entire gross revenue from the tax via the rebate, and proportionally adjust other taxes in the model to achieve the same level of long-run debt.

By holding the real carbon tax rate constant over time, these simulations differ from most policy proposals for a carbon tax, which typically involve a tax rate that rises over time (at a rate faster than inflation). We do this for simplicity: it is easier to understand and to explain the effects of a constant carbon tax rate than the effects of a carbon tax rate that changes over time. Similarly, many proposals would announce the carbon tax prior to when it is imposed, so that the economy has some time to adjust before the policy takes effect.[5] Again, for simplicity, we assume that entire policy change is implemented immediately, without any pre-announcement.

## IV. MODEL

Our modeling framework involves the calculation of an intertemporal general equilibrium with overlapping generations and perfect foresight. Results from this equilibrium are refracted onto income-level subgroups of the population using a microsimulation model.

### A. General Equilibrium Modeling

The general equilibrium modeling uses a dynamic overlapping generation (OLG) model of the U.S. economy. That model was first developed in Carbone, Morgenstern, and Williams (2012), which provides full details. We provide a brief overview below.

The model describes how the U.S. economy evolves over time. The model combines two key components: overlapping generations (important for modeling the dynamics of growth and decision-making over time) and multi-sector production (important for modeling tradeoffs across different industries). Following the standard overlapping-generations approach, the model introduces a new generation in each five-year model period, which then makes lifecycle consumption and savings decisions for its 55-year

[4] Indexing transfers for inflation would imply holding real transfers constant in each period, not just in net present value. Several papers have shown that indexing of existing transfer programs substantially reduces the regressivity of carbon pricing (Parry and Williams, 2010; Blonz, Burtraw, and Walls, 2012; Dinan, 2012; Fullerton, Heutel, and Metcalf, 2012).

[5] However, Williams (2012) suggests that it is more efficient not to pre-announce environmental policy.

economic lifetime,[6] and then exits the model. Thus, at any point in time, there are 11 active generations at different stages of the lifecycle. Each generation is modeled as a single representative household. Those representative households are scaled such that each generation represents a larger number of people than the previous generation, based on a constant exogenous population growth rate of 1 percent.

Each of the representative households derives income from labor, capital, natural resources, and government transfers. Each household makes savings, consumption, and labor-supply decisions in order to maximize the discounted sum of utility over its lifespan, subject to its budget constraint. Household utility in any given period is a constant-elasticity-of-substitution (CES) function of consumption ($c_{gt}$) and leisure ($l_{gt}$). Aggregate household choices determine the levels of national income, consumption, saving, and investment levels in the economy, which in turn determine how the aggregate capital stock and the economy grow over time.

Production occurs in nineteen competitive, constant-returns-to-scale industries including 16 intermediate goods: 15 energy and energy-intensive industries (those most directly affected by a carbon tax) that are taken directly from the disaggregated Global Trade Analysis Project (GTAP) 7.1 dataset, and one composite of all other intermediate goods. The remaining industries produce government services, investment, and a household consumption good.[7]

Firms in each industry combine basic factors of production (physical capital, labor, and natural resources) with intermediate inputs. Each industry has a nested CES production function, with the nesting implying that substitution among energy goods is easier than substitution between energy and other inputs. We assume a constant, exogenous 2 percent rate of labor-augmenting technological change in every industry. Goods can also be imported or exported. We assume international prices are fixed (i.e., the small open economy assumption), and that domestic and international varieties of each good are imperfect substitutes for each other.

The model also includes a representation of government taxes, spending, and transfers. This government sector represents an aggregate of federal, state, and local governments. The government has taxes on labor income (representing federal and state income taxes on labor income, plus social security payroll taxes), capital income (federal and state corporate income taxes and personal income taxes on capital income), consumption spending (primarily state sales taxes), and carbon emissions (with the carbon tax initially set to zero).

Tax revenue finances government services and transfer payments to households. In the benchmark case (i.e., the case without any policy changes), spending on government services and on transfer payments grows over time at the same rate as GDP. Each policy

[6] This 55-year period is meant to approximate the typical time from first entry into the labor force until death.

[7] Note that this is equivalent to having government and households directly purchase the "intermediate" goods (and indeed, when we decompose incidence by income and geography, we take account of different patterns of consumption good purchases across income groups and states).

case we consider keeps the cumulative real net present value of government services and transfer payments constant, but the value in any given year may vary slightly from the benchmark.

The government may run a deficit or surplus in any given time period. In the benchmark case, the government runs a persistent budget deficit of approximately 3 percent of GDP forever (based on long-term budget projections from CBO (2012)). This implies an ever-increasing level of national debt. Each policy case we consider may have changes in budget deficits in the short run, but maintains the long-run path of national debt the same as in the benchmark case.

## B. Decomposition of Incidence by Commodity and Income Source

The OLG model estimates the aggregate price and quantity changes caused by a policy and the welfare effects on different generations. The incidence model refracts that national result onto the distribution of households across income groups. Changes to welfare due to the policy can be decomposed according to changes in welfare stemming from changes in the price and quantity of commodities (which we approximate using consumer surplus) and welfare changes stemming from changes in income (which we approximate using producer surplus).[8] The general equilibrium results from the OLG model are:

1. In the benchmark (no policy) case, expenditures ($x_i$) for 17 commodity goods ($i$) and incomes ($y_j$) from eight income sources ($j$), for each time period.
2. In each policy case, the percentage change in price and quantity for each of these commodities and income sources. For government transfers and the potential rebate under a carbon tax, the model provides the absolute change in income.

Assuming the demand curve is linear in the relevant range of changes in price and quantity, the equilibrium representation of changes to consumer surplus for a given commodity $i$ is the change in consumer surplus, calculated as

$$(1) \quad \Delta cs_i = -\left(1 + \frac{1}{2}\frac{\Delta q_i}{q_i}\right)\frac{\Delta p_i}{p_i} x_i .$$

A partial equilibrium demand curve holds the prices of other goods and sources of income constant. Thus a consumer surplus calculation based on such a demand curve would omit the welfare implications of interactions between price changes for different

[8] These represent very close approximations: in each policy case, summing producer and consumer surplus changes for the whole economy yields a result that is nearly identical to the sum of equivalent variations across the representative households in the OLG model. West and Williams (2004) find that consumer surplus provides a good approximation to equivalent variation when calculating incidence.

goods, or between price changes and income changes. However, we take the quantity changes from the OLG model equilibrium results. Thus, the consumer surplus calculation for good $i$ implicitly accounts for all of the interactions among the effects of the price change for good $i$ and changes in income and in the prices of other goods.

Analogously, assuming the supply curve is linear in the relevant range of changes in price and quantity, the equilibrium representation of changes to producer surplus for a given income source $j$ is the change in producer surplus for each income source, calculated as

$$(2) \quad \Delta ps_j = \left(1 + \frac{1}{2}\frac{\Delta q_j}{q_j}\right)\frac{\Delta p_j}{p_j} y_j .$$

As with the consumer surplus calculation, the producer surplus calculation takes quantity changes from the OLG model equilibrium, and thus implicitly accounts for interactions of different prices.

The quantity of natural resources is exogenously fixed in the model, and thus there is no quantity change for natural resource income. Two sources of income, government transfers and potential rebates from a carbon tax, have no marginal cost schedule and no change in price, and thus the change in "producer surplus" for that income source is simply equal to the change in income from that source.

Incidence is calculated as the sum of the changes in consumer and producer surplus

$$(3) \quad \Delta W_i = \sum_i \Delta cs_i + \sum_j \Delta ps_j .$$

### C. Calculating Welfare Effects Caused by Changes in Consumer Good Prices

We index changes in welfare, $\Delta W^k$, where $k$ is an index of income quintiles. For all goods we assume that changes to prices and quantities due to the policy are constant across income quintiles. However, the initial level of expenditures of each commodity and income by source is different ($x_i^k$ and $y_j^k$). Using the changes to consumer and producer surplus indicated previously, we apply the following equations to predict welfare changes across income groups

$$(4) \quad \Delta cs_i^k = \Delta cs_i \left(\frac{x_i^k}{x_i}\right) \text{ and } \Delta ps_j^k = \Delta ps_j \left(\frac{y_j^k}{y_j}\right).$$

Incidence is calculated as

$$(5) \quad \Delta W^k = \sum_i \Delta cs_i^k + \sum_j \Delta ps_j^k .$$

The data needed from outside the OLG model are the percentages of expenditures and income by quintile ($x_i^k/x_i$ and $y_j^k/y_j$).

The OLG model provides price and quantity changes (from which consumer surplus changes are calculated) for 17 commodities ($i$), which are higher level categorizations of the 57 commodity sectors in the GTAP database, and do not represent final consumption goods. We use a transformation matrix (Elliott and Fullerton, 2014) to match consumption goods to the commodity goods in the Bureau of Economic Analysis Personal Consumption Expenditures (PCE) categories. For example, ferrous metal is a commodity that corresponds to the consumption goods furnishings, appliances, and autos.

To represent expenditure across income quintiles, we use the Consumer Expenditure Survey (CEX), conducted by the Bureau of Labor Statistics (BLS), which is a quarterly survey of randomly sampled households from 91 geographically contiguous primary sampling areas across the United States. The CEX also includes data on household income, and income quintiles are comprised by calculating individual income as household income divided by the square root of the number of members of the household, and creating five groups with an equal number of individuals. Goods are organized by Universal Classification Codes (UCC) in the CEX data, which are transformed into PCE categories to be comparable to the data from the OLG model data that has also been transformed to PCE categories. We describe the proportion of spending across UCC consumption goods $u$ and income quintiles $k$ as $\tilde{x}_u^k / \tilde{x}_u$.

## D. Calculating Effects on Household Income

There are seven income goods in the OLG model: five sources of asset income (including capital and multiple types of natural resources), plus transfers and labor. We assume that the mix of the five sources of asset income is constant across households, as most of these assets are held as securities in diverse portfolios. Consequently we use estimates of the proportion of income by three sources: capital, transfers, and labor. The price changes for income sources from the OLG model are tax-inclusive (e.g., the price change for labor is the change in the after-tax wage from the OLG model equilibrium), but we allocate this across households based on pre-tax income shares (thus implicitly assuming that any tax rate change is an equal percentage-point tax rate change at all income levels).

To develop estimates of income by quintile, we rely on the Congressional Budget Office (CBO, 2010), which provides annual estimates. These estimates are built on one administrative record, the Internal Revenue Service's Statistics of Income (SOI), and one survey, the Census Bureau Current Population Survey (CPS). The SOI does not record information from low-income households (who do not pay federal income taxes) and the CPS underreports income from capital (which is often the case in self-reported surveys) and lacks information on high-income households, so the CBO estimates attempt to correct for these factors. CBO provides before-tax income estimates by quintile (calculated in the same way we calculate quintiles in the CEX data) for labor and transfer income. We sum CBO estimates (identified with a double bar) for 2010 of capital gains, business income, capital income, and other income (which is mostly retirement income) by quintile to create our capital income estimates. The change in producer surplus across quintiles is calculated such that

$$(6) \quad \Delta ps_j^k = \Delta ps_j \times \frac{\bar{\bar{y}}_j^k}{\bar{\bar{y}}_j}.$$

## V. RESULTS ACROSS INCOME GROUPS

The results from the OLG model describe an intertemporal equilibrium affecting overlapping generations. But the focus here is on the short-run effect of the policy — that is, how households are affected by the immediate short-term effects of the policy on prices, wages, and returns to capital. However, note that these immediate effects come from the intertemporal equilibrium (and thus implicitly anticipate changes that will occur in the future).

All of the results we report omit the environmental benefits from the carbon tax stemming from reduced emissions of greenhouse gases and changes in conventional air pollutants that occur. The reduction in emissions is very similar (though not identical) across the different policy cases, so not accounting for benefits will not significantly affect the relative attractiveness of different policy options (either for the overall mean or for particular subgroups).

The normalization used in the CGE model implies that the price of the average consumer good remains constant. This normalization does not affect the incidence calculation (Fullerton and Metcalf (2002) provide a discussion of price normalizations and tax incidence). It allows us to interpret the welfare changes in real terms — that is, all consumption good price changes should be interpreted as real price changes (i.e., price changes relative to the average consumer good), and income changes should be interpreted as changes in real after-tax wages, returns to capital, or government transfers.

Results are reported as the change in welfare measured in dollars per household. Table 1 provides an overview of the effects on the mean household at the national level in the first year of the climate policy.[9] In the case that recycles tax revenue to reduce the capital income tax the cost is \$291, reducing the labor income tax yields a cost of \$407, and rebating the revenue in equal (lump sum) payments yields a cost of \$866, more than twice as much as labor tax recycling. This order is expected and generally agrees with what we know about tax efficiency. There is no policy that yields a "double dividend" for the mean household. In other words, ignoring the benefits from reduced carbon emissions, no policy is preferable to the status quo.

Effects on the mean household overall differ substantially from effects on the middle (third) income quintile (which approximates the median). The order of preference is completely reversed for the middle quintile, compared to the mean. The effects on the

[9] We use the "mean household" as shorthand for the mean across all households. Similarly, when we report effects on a quintile, these are the means of those effects across all households in that quintile.

**Table 1**

Change in Welfare by Household
$(2012)

| | Middle Quintile Household ($) | | | Mean Household ($) | | |
|---|---|---|---|---|---|---|
| | Capital Recycling | Labor Recycling | Lump Sum Rebate | Capital Recycling | Labor Recycling | Lump Sum Rebate |
| *Energy Goods* | *–521* | *–534* | *–531* | *–530* | *–543* | *–540* |
| Electricity | –169 | –174 | –166 | –175 | –180 | –172 |
| Fuel oil & other | –30 | –31 | –31 | –32 | –33 | –34 |
| Gasoline | –236 | –242 | –248 | –233 | –239 | –245 |
| Natural gas | –86 | –87 | –86 | –90 | –91 | –89 |
| *Other Goods* | *476* | *490* | *481* | *529* | *543* | *539* |
| *Sources of Income* | *–489* | *–120* | *329* | *–290* | *–406* | *–865* |
| Capital income | 215 | –166 | –527 | 556 | –429 | –1,361 |
| Labor income | –448 | –25 | –834 | –638 | –35 | –1,188 |
| Rebate | – | – | 1,591 | – | – | 1,603 |
| Transfer income | –257 | 71 | 99 | –208 | 58 | 80 |
| *Total* | *–534* | *–163* | *279* | *–291* | *–407* | *–866* |

Note: Welfare changes omit environmental benefits of the carbon tax.

middle-quintile household range from a cost of $534 with a capital tax swap, a cost of $163 with a labor tax swap, to a gain of $279 with rebates. Rebates are the best choice for the middle quintile, but the worst choice for the mean household. The labor income tax swap is the second choice for both the middle-quintile and the mean household.

The policies do not vary substantially in their effect on the welfare loss associated with direct energy goods. For both the middle-quintile and mean households, the spread is $13. The biggest welfare loss among energy goods comes from gasoline consumption. On the other hand, the biggest difference across income groups stems from the effect on sources of income. This difference results in part because of the substantial difference between mean and median income (annual income for the mean household is $127,591 compared to $93,174 for the middle quintile) and partly because of the difference in income sources (capital income represents a larger share of aggregate income overall than it does for the middle quintile).

The effects across the income distribution appear in Table 2, which reports the impact per household as a percentage of annual income for the nation and for each income quintile. Three quintiles prefer the lump sum rebate and these households are strictly winners in this case. The only quintile that prefers the labor tax recycling case is the fourth quintile, although in no case does the fourth quintile strictly benefit (receive a double dividend). The only quintile that prefers the capital tax swap is the fifth quintile.

The welfare loss as a percentage of income is shown in the first panel of Figure 1 and in absolute terms (dollars per household) in the second panel. In general terms, the labor tax swap would be the second choice of all the quintiles except for the fourth, who prefer it most of all. The labor tax swap is never the least favorite recycling option for any income group.

Except for the difference between the lowest income quintile and the second lowest quintile under the labor tax swap, the direction of changes in welfare align across all quintiles for all the policies. For example, under the capital tax swap, the welfare cost as a percentage of income is strictly decreasing with income and is least for the highest income quintile, while for the labor tax swap and the lump sum rebate it is strictly increasing with income.

In percentage terms, Figure 1 shows a majority of the quintiles receive a double dividend in the lump sum rebate case, the first quintile gains more than the fifth quintile loses, and the second quintile gains more than the fourth quintile loses. Nonetheless, the mean household experiences a loss because 69.3 percent of income is concentrated in the top two quintiles.

We summarize the results with the observation that a policy maker choosing among these three options would pick the lump sum rebate if she cared most about what the majority would vote for, or to reduce inequality. The policy maker would pick the labor tax swap if she wanted to have the most even effect on all income quintiles, or to avoid the worst case scenario for each income quintile. The policy maker would pick the capital income tax swap if she cared most about total welfare.[10]

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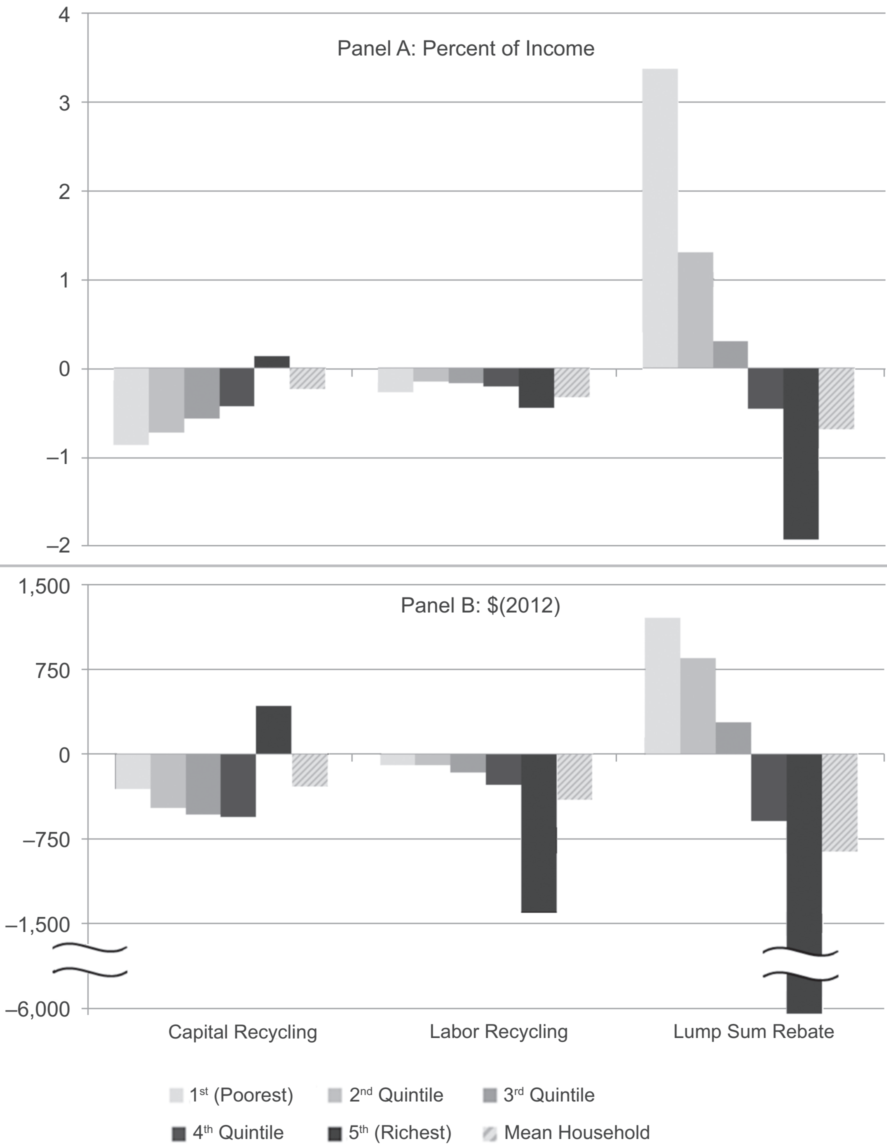
[10] Of course, real-world policy makers have a much broader set of options. They could divide the revenue among two or more of these recycling options (e.g., devoting some revenue to capital tax cuts to boost efficiency and some to rebates in order to limit the regressivity of the policy). This would yield an outcome that would be a weighted average of the results here. Or they could use more complex revenue-recycling approaches tailored to a particular goal (e.g., if the goal is to reduce inequality, a means-tested transfer would be more effective than the uniform transfer we model).

**Table 2**
Change in Welfare by Quintile
(Percentage of Income)

| | 1st (Poorest) | 2nd Quintile | 3rd Quintile | 4th Quintile | 5th (Richest) | U.S. Total |
|---|---|---|---|---|---|---|
| *Capital Recycling* | –0.87 | –0.73 | –0.57 | –0.43 | 0.14 | –0.23 |
| Energy goods | –1.04 | –0.70 | –0.56 | –0.46 | –0.22 | –0.42 |
| Electricity | –0.41 | –0.25 | –0.18 | –0.14 | –0.07 | –0.14 |
| Fuel oil & other | –0.06 | –0.04 | –0.03 | –0.03 | –0.02 | –0.03 |
| Gasoline | –0.40 | –0.29 | –0.25 | –0.22 | –0.10 | –0.18 |
| Natural gas | –0.17 | –0.12 | –0.09 | –0.07 | –0.04 | –0.07 |
| Other Goods | 0.81 | 0.59 | 0.51 | 0.46 | 0.29 | 0.41 |
| Sources of income | –0.64 | –0.63 | –0.53 | –0.43 | 0.08 | –0.23 |
| Capital income | 0.15 | 0.16 | 0.23 | 0.30 | 0.64 | 0.44 |
| Labor income | –0.40 | –0.42 | –0.48 | –0.55 | –0.51 | –0.50 |
| Rebate | – | – | – | – | – | – |
| Transfer income | –0.39 | –0.37 | –0.28 | –0.17 | –0.06 | –0.16 |
| *Labor Recycling* | –0.28 | –0.15 | –0.18 | –0.21 | –0.45 | –0.32 |
| Energy goods | –1.06 | –0.72 | –0.57 | –0.47 | –0.23 | –0.43 |
| Electricity | –0.42 | –0.25 | –0.19 | –0.15 | –0.07 | –0.14 |
| Fuel oil & other | –0.06 | –0.04 | –0.03 | –0.03 | –0.02 | –0.03 |
| Gasoline | –0.41 | –0.30 | –0.26 | –0.22 | –0.10 | –0.19 |
| Natural gas | –0.17 | –0.12 | –0.09 | –0.08 | –0.04 | –0.07 |
| Other goods | 0.82 | 0.61 | 0.53 | 0.47 | 0.29 | 0.43 |
| Sources of income | –0.03 | –0.04 | –0.13 | –0.21 | –0.51 | –0.32 |
| Capital income | –0.12 | –0.12 | –0.18 | –0.23 | –0.50 | –0.34 |
| Labor income | –0.02 | –0.02 | –0.03 | –0.03 | –0.03 | –0.03 |
| Rebate | – | – | – | – | – | – |
| Transfer income | 0.11 | 0.10 | 0.08 | 0.05 | 0.02 | 0.05 |
| *Lump Sum Rebate* | 3.36 | 1.29 | 0.30 | –0.46 | –1.93 | –0.68 |
| Energy goods | –1.05 | –0.71 | –0.57 | –0.47 | –0.23 | –0.42 |
| Electricity | –0.40 | –0.24 | –0.18 | –0.14 | –0.07 | –0.13 |
| Fuel oil & other | –0.06 | –0.04 | –0.03 | –0.03 | –0.02 | –0.03 |
| Gasoline | –0.42 | –0.31 | –0.27 | –0.23 | –0.11 | –0.19 |
| Natural gas | –0.17 | –0.12 | –0.09 | –0.07 | –0.04 | –0.07 |
| Other goods | 0.84 | 0.61 | 0.52 | 0.46 | 0.29 | 0.42 |
| Sources of income | 3.57 | 1.39 | 0.35 | –0.45 | –2.00 | –0.68 |
| Capital income | –0.37 | –0.39 | –0.57 | –0.73 | –1.58 | –1.07 |
| Labor income | –0.75 | –0.78 | –0.90 | –1.03 | –0.95 | –0.93 |
| Rebate | 4.55 | 2.42 | 1.71 | 1.24 | 0.51 | 1.26 |
| Transfer income | 0.15 | 0.14 | 0.11 | 0.07 | 0.02 | 0.06 |

Note: Welfare changes omit environmental benefits of the carbon tax.

**Figure 1**
Welfare Change per Household by Quintile




Note: Welfare changes omit environmental benefits of the carbon tax.

Presenting welfare loss in dollar terms rather than as a percentage of income provides a different perspective (the second panel of Figure 1). This perspective highlights that the lump sum rebate policy results in average losses of just over $6,000 a year for top-quintile households, which is much more in absolute terms than the $1,202 and $846 average gains for first- and second-quintile households under the policy. From this perspective the labor income recycling policy also looks less attractive even than the capital recycling policy. The labor income recycling policy imposes costs that grow over the income distribution with the greatest costs incurred by the fifth income quintile. Moreover, the poorest quintile is no longer the worst off under the capital income recycling policy; rather, the policy imposes growing losses over the income distribution except for the fifth quintile, where gains are concentrated.

The components of welfare loss across the income quintiles are evident in Table 2 as a percentage of income. Similar effects are observed for direct energy goods and other consumer goods over the three policies. The impact as a percentage of income of the changes in welfare associated with energy goods is regressive across quintiles largely because expenditure on energy goods is relatively constant across quintiles, but the denominator, income, is not constant.

The welfare changes associated with consumption of other (nonenergy) goods consistently result in increases in welfare. This is a result of the price normalization, which holds the price of the average consumer good constant, so that the price increase for energy goods means that the relative price of non-energy goods falls. Other consumption is more evenly spread across quintiles than is income. However, since expenditures on energy goods are more equal across quintiles than expenditures on other goods, the regressive nature of the welfare loss from direct energy goods overwhelms the progressive nature of the welfare gains from other goods. Therefore, before considering welfare changes from income, we observe that poorer quintiles are consistently worse off because energy consumption comprises a larger proportion of their total consumption.

The big differences between the outcomes of the policies stem from the welfare changes due to changes in the sources of income. Under the capital tax swap there is a significant welfare gain to capital income and a significant welfare loss to labor income. Because capital income is more concentrated in the top quintile than labor income, this results in a regressive net effect from these sources. The loss in transfer income also is a contributing factor to the regressive outcome. As noted previously, we hold the cumulative real net present value of transfers constant for all policy changes, but transfers may vary in any given period. Over the longer term, capital tax recycling leads to slightly lower consumer prices, implying lower nominal transfers. But because that effect on consumer prices takes several periods to appear, the reduction in transfers implies a small drop in real transfers initially.

In the labor income tax swap, the large welfare loss to labor income is almost completely erased, as the cut in labor tax rates means that real after-tax wages remain roughly constant (recall that the price normalization means that all changes should be interpreted in real terms). But real after-tax returns on capital fall substantially in this case. Because capital income is concentrated in higher-income quintiles, this produces a progressive effect overall.

Under the lump sum rebate there are large reductions in after-tax wages and returns to capital, producing substantial welfare losses from those sources. The rebate income is equal across quintiles in dollar terms, but represents a much larger proportion of income for the poorer quintiles than it does for the top quintile. All of these changes in income have very progressive effects, especially the rebate itself, which is why the lump sum rebate policy is the most uneven in its effects and the most progressive policy option.

## VI. CONCLUSION

Our results show that the effects of a carbon tax on energy prices are somewhat regressive, consistent with earlier studies. However, we also find that the use of revenue can easily offset that effect. As expected, the mean household does the worst under a lump sum rebate, but we find that this policy will create a double dividend for the bottom three quintiles of the U.S. population in the first year. This finding should be particularly interesting to policy makers who are interested both in internalizing the price of carbon and reducing inequality. And if households vote based on their self-interest, such a policy could be politically quite popular.

In contrast, recycling revenue to reduce capital taxes is the most efficient policy, but we caution that it makes carbon pricing, which is already regressive, even more so (at least in the short run). Meanwhile, using carbon tax revenue to reduce taxes on labor is a clear middle of the road option: While it is less efficient than recycling revenue to cut capital income taxes, the efficiency difference is relatively modest, and cutting labor income taxes offsets some of the natural regressivity of a carbon tax. The distributional effects are close to even across the income distribution, when measured as a percentage of income, which could have political advantages. In contrast, if the rebate policy is viewed as an attempt to reduce inequality, this might introduce a controversial policy sub-plot to an already polarized debate.

Our findings are necessarily limited by the detail available in the OLG model. One important limitation is the assumption of a single national labor market, with no differentiation by skill level or region. Workers in different locations and at different skill levels are likely to be affected by and respond differently to a carbon tax, and our model misses those differences.

Finally, this paper looks only at incidence across incomes and only at incidence in the short run. In ongoing work we examine the distributional effects of these three policies by state and region. In related work in progress, we look at the transitional incidence of the three policies by quintile and geography over a longer time frame to trace incidence through the transition to a new long-run equilibrium. This may be particularly important for the capital tax recycling case where the initial benefit of the capital tax cut goes to owners of capital, but over time as the capital stock grows, some of that benefit is passed through to workers and consumers in the form of higher wages and lower consumer prices.

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## DISCLOSURES

The authors have no financial arrangements that might give rise to conflicts of interest with respect to the research reported in this paper.

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